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1.3 Taxonomy and Systematics

Taxonomy (G. *taxis*- arrangement ; *nomos*- law) is the science of arrangement of living organisms along with classification, description, identification, and naming of organisms which includes all flora and fauna including microorganisms of the world. The word taxonomy was coined by **Augustin Pyramus de Candole (1813).** Taxonomy is a theoretical study of classification with well defined principles, rules and procedures. **Aristotle** is called the father of taxonomy (classical) and **Carolus Linnaeus** is the father of modern taxonomy.

Systematics (G. *System*/sequence)

The objectives of taxonomy and systematics are very similar; their goal is to classify organisms with stipulated rules. The main criteria of systematics is identifying, describing, naming, arranging, preserving and documenting the organisms. Apart from the above said features, evolutionary history of the species and the environmental adaptations and interrelationship between species are also being investigated in systematics.



Carolus Linnaeus is the father of modern taxonomy, which is the system of classifying and naming organisms. One of his contributions was the development of a hierarchical system of classification of nature. Today, this system includes eight taxa: domain, kingdom, phylum, class, order, family, genus, and species.



History of Classification

Early classification of organisms were based on only two criteria, beneficial or harmful animals. An ancient classification system recognized 5 animal groups - domestic, wild, creeping, flying and sea animals. Initially the classification was based on organism's fundamental characteristics such as the habitat and morphology only.

Aristotle (384 to 322 BC), was the first to classify all animals in his **History of Animals** (*Historia Animalium* in Latin). He attempted a basic classification of all living organisms into Plants and Animals. Animals were classified based on locomotion; walking (terrestrial), flying (aerial) and swimming (aquatic). Based on the presence or absence of red blood he classified the animals into two as *Enaima* with blood and those without blood as *Anaima*.

Aristotle's classification system had limitations and many organisms were not fitting into his classification. For example, the tadpoles of frogs are born in water and have gills but when they metamorphosed into adult frogs they have lungs and can live both in water and on land. How to classify frogs and where to place them? Aristotle classified organisms based on locomotion, hence, birds, bats, and flying insects were grouped together just by observing one single characteristic feature, the flying ability. On the contrary to the above said